

CANBISU POWDER 6mg/g

Ingredient(s):

Each g contains :
Ivermectin 6mg

Preservative(s):

Each g contains :
Sodium Benzoate 1mg

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

Ivermectin enhances the release of gamma amino butyric acid (GABA) at pre-synaptic neurons. GABA acts as an inhibitory neurotransmitter and blocks the post-synaptic stimulation of the adjacent neuron in nematodes or the muscle fiber in arthropods. By stimulating the release of GABA, ivermectin causes paralysis of the parasite and eventual death.

In monogastric animals, ivermectin is up to 95% absorbed after oral administration. Ruminants only absorb 1/4 to 1/3 of a dose due to inactivation of the drug in the rumen. While there is greater bioavailability after SC administration, absorption after oral dosing is more rapid than SC. It has been reported that ivermectin's bioavailability is lower in cats than in dogs.

Ivermectin is well distributed to most tissue, but does not readily penetrate into the CSF. Collie-breed dogs apparently allow more ivermectin into the CNS than other breeds/species.

Ivermectin has a long terminal half-life in most species. It is metabolized in the liver via oxidative pathways and is primarily excreted in the feces. Less than 5% of the drug (as parent compound or metabolites) is excreted in the urine.

Pharmacokinetic parameters of ivermectin have been reported for various species:

Species	Bioavailability (F)	Volume of distribution (Vd) (L/kg)	T1/2 (terminal) (in days)	Total Body Clearance (L/kg/day)
Cattle		0.45~2.4	2~3	0.79
Dogs	0.95	2.4	2	
Swine		4	0.5	
Sheep	1.0 intra-abomasal 0.251 intra-ruminal			

Indication(s):

The product effectively controls the following parasites of swine when administered in the feed to provide the recommended dose level of 0.1mg ivermectin per kg bodyweight daily for 7 days:

Gastrointestinal roundworms:

Ascaris suum (adults and L4), *Hyostrongylus rubidus* (adults and L4), *Oesophagostomum spp.* (adults and L4), *Strongyloides ransomi* (adults)*

Lungworms:

Metastrongylus spp. (adults)

Lice:

Haematopinus suis

Mange mites:

Sarcoptes scabiei var. suis

*Given to pregnant sows before farrowing, it effectively controls transmission via milk of *S.ransomi* to piglets.

Target Species

Swine.

Dosage and Administration:

16.7 mg Canbisu/kg bodyweight fed daily for 7 consecutive days.

To ensure thorough dispersion of the product it should first be mixed with a suitable quantity of feed ingredients before incorporation into mix. Only to be mixed in dry feed.

Growing Pigs

BW less than 40 kg: 333 g Canbisu/tonne of final feed.

BW 40 kg and above: 400 g Canbisu/tonne of final feed.

Mix thoroughly in the finished feed and feed continuously as the only ration for 7 consecutive days.

Adult Pigs

BW over 100 kg: 1.67 kg Canbisu/tonne feed given at a feeding rate of 1 kg /100kg BW daily for 7 days. Where medicated feed is to be fed as part of the ration, it is recommended that the medicated feed is fed first.

RECOMMENDED TREATMENT PROGRAMME

Growing Pigs: It is recommended that the in-feed parasite control programme should begin with treatment of all growing pigs for 7 consecutive days.

Breeding animals: Feed with medicated feed for 7 consecutive days. At the time of initiating any parasite control programme, it is important to treat all animals in the herd. After the initial treatment, use the premix regularly as follows:

Sows: Treat 14-21 days, prior to farrowing, to minimize infection of piglets.

Gilts: Treat 14-21 days prior to breeding. Treat 14-21 days prior to farrowing.

Boars: Treat at least 2 times per year. Frequency and need for treatments are dependent upon parasite exposure.

Oral via feed mixing.

Contraindications:

- 1) Ivermectin is not recommended to use in foals less than 4 months old, as safety of the drug in animals this young has not been firmly established. However, foals less than 30 days of age have tolerated doses as high as 1mg/kg without symptoms of toxicity.
- 2) Ivermectin is not recommended for use in puppies less than 6 weeks old. Most clinicians feel that ivermectin should not be used in Collies or Collie-mix breeds at the doses specified for treating microfilaria or other parasites unless alternative therapies are unavailable. After receiving heartworm prophylaxis doses, it is recommended to observe Collie-breeds for at least 8 hours after administration.

Precaution(s) / Warning(s):

This product is for use in swine only. Should not be used for other animal species.

Special precautions to be taken by the person administering the medicinal product to the animals

Do not smoke, drink or eat while handling the product.

Wash hands after use.

Mixing of the product with feed must take place in a well ventilated area. Avoid contact with skin and eyes.

In case of accidental contact, wash the affected area thoroughly with clean running water. If eye irritation persists, seek medical advice.

Interaction with Other Medicaments

No drug interaction listed.

Pregnancy and Lactation

- 1) Because milk withdrawal times have not been established, the drug is not approved for use in lactating dairy animals or females of breeding age.
- 2) Ivermectin is considered to be safe to use during pregnancy. Reproductive studies performed in dogs, horses, cattle and swine have not demonstrated adverse effects to fetuses. Reproductive performance in male animals is also apparently unaltered.

Side Effect(s) / Adverse Reaction(s):

- 1) In horses, swelling and pruritus at the ventral mid-line can be seen approximately 24 hours after ivermectin administration due to a hypersensitivity reaction to dead *Onchocerca* sp. microfilaria. The reaction is preventable by administering a glucocorticoid just prior to, and for 1~2 days after ivermectin. If untreated, swelling usually subsides within 7~10 days and pruritus will resolve within 3 weeks.
- 2) Dogs may exhibit a shock-like reaction when ivermectin is used as a microfilaricide, presumably due to a reaction associated with the dying microfilaria.
- 3) When used to treat *Hypoderma bovis* larva (Cattle grubs) in cattle, ivermectin can induce serious adverse effects by killing the larva when they are in vital areas. Larva killed in the vertebral canal can cause paralysis and staggering. Larva killed around the gullet can induce salivation and bloat. Cattle may also experience discomfort or transient swelling at the injection site.
- 4) In mice and rats, ivermectin may cause neurologic toxicity at doses slightly more than usually prescribed (less than 0.5mg/kg).
- 5) In birds, death, lethargy or anorexia may be seen. Orange-cheeked Waxbill Finches and budgerigars may be more sensitive to ivermectin than other species.

Environmental Property

Studies indicate that when ivermectin comes in contact with the soil, it readily and tightly binds to the soil and becomes inactive over time. Free ivermectin may adversely affect fish and certain water-borne organisms on which they feed. Do not permit water runoff from feedlots or production sites to enter lakes, streams, or ponds. Do not contaminate water by direct application or by the improper disposal of drug container. Dispose of containers in an approved landfill or by incineration.

Symptoms and Treatment for Overdosage, and Antidote(s):

- 1) In horses, doses of 1.8mg/kg (9 times recommended dose) PO did not produce symptoms of toxicity, but doses of 2mg/kg caused symptoms of visual impairment, depression and ataxia.
- 2) In cattle, toxic effects generally do not appear until dosages of 30 times those recommended are injected. At 8mg/kg, cattle showed symptoms of ataxia, listless, and occasionally death.
- 3) Sheep showed symptoms of ataxia and depression at ivermectin doses of 4mg/kg.
- 4) Swine showed symptoms of toxicosis (lethargy, ataxia, tremors, lateral recumbency, and mydriasis) at doses of 30mg/kg. Neonatal pigs may be more susceptible to ivermectin overdoses, presumably due to more permeable blood-brain barrier.
- 5) In dogs, symptoms of acute toxicity rarely occur at single dosages of 1mg/kg or less. At 2.5mg/kg mydriasis occurs, and at 5mg/kg tremors occurs. At doses of 10mg/kg, severe tremors and ataxia are seen. Deaths occurred when dosages exceeded 40mg/kg, but the LD50 is 80mg/kg. Dogs (Beagles) receiving 0.5mg/kg PO for 14 weeks developed no signs of toxicity, but at 1~2mg/kg for the same time period, developed mydriasis and had some weight decreases. Half of the dogs receiving 2mg/kg/day for 14 weeks developed symptoms of depression, tremors, ataxia, anorexia, and dehydration. The collie breed appears to be more sensitive to the toxic effects of ivermectin than other canine breeds. This may be due to a more permeable blood-brain barrier to the drug accumulation in the CNS of this breed.
- 6) Dogs who receive an overdosage of ivermectin or develop signs of acute toxicity (CNS effects, GI, cardiovascular) should receive supportive and symptomatic therapy. Emptying the gut should be considered for recent massive oral ingestions in dogs or cats.
- 7) Acute toxic symptoms associated with massive overdoses in cats will appear within 10 hours of ingestion. Symptoms may include agitation, vocalization, anorexia, mydriasis, rear limb paresis, tremors, and disorientation. Blindness, head-pressing, wall-climbing, absence of oculomotor menace reflex, and a slow and incomplete response to pupillary light may also be seen. Neurological symptoms usually diminish over several days and most animals completely recover within 2~4 weeks. Symptomatic and supportive cares are recommended.

Shelf-Life:

3 years.

Storage Condition(s):

Store below 30°C.

Product Description and Packing(s):

A white powder.

Al. Pouch of 333g

For food producing animals product:

Withdrawal Periods

Pigs (over 100kg) : 12 days

Pigs (under 100kg) : 3 days

Maximum Residual Limit (MRL):

All mammalian food producing species : Fat 100 µg/kg, liver 100 µg/kg, kidney 30 µg/kg.

Not for use in animals producing milk for human consumption.

Disposal:

Dispose of any unused product or containers in accordance with guidance of the local waste regulation authority.

**Made by YSP**

Manufacturer:

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